RESEARCH-ARTICLE

Equality Engine: Fostering Critical Machine Learning Bias Literacy Through a Transformational Game

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Machine Learning (ML) is now integrated from everyday technologies to sophisticated infrastructures, providing fast, efficient, and scalable decision-making services, with increasing evidence of ML perpetuating invisible harms and biases. The hidden and socio-technical nature of ML biases can make them difficult to detect and prevent without proper literacy. To investigate this, we developed a novel online multiplayer board game Equality Engine, where players learn about various ML biases and debiasing techniques. We conducted a mixed-method formative playtest case study to solicit feedback from post-secondary students (N = 12) with a range of ML experience. We found that students' self-reported ML bias-debias knowledge improved significantly after playing the game. The game was perceived as easy to use because of the social interaction and immersion the game enabled. Students would also use the game in the future because of the self-reported knowledge gained from the game. Although these positive results may be influenced by measurement bias, our study contributes to the design of an approachable game, which not only facilitates exposure, collaboration, and opportunities for critical reflection on ML biases but also provides recommendations for future game designs that can facilitate ML ethics discourse and literacy among a broader audience.

CCS Concepts: • **Applied computing → Education**; • **Social and professional topics → Computing literacy**; • **Human-centered computing** → *Empirical studies in HCI*; • **Computing methodologies** → *Machine learning*.

Additional Key Words and Phrases: Machine Learning, ML Biases, ML ethics, AI/ML literacy, game design, transformational games

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1 Introduction

Artificial Intelligence (AI) and Machine Learning (ML) driven systems are deployed in every aspect of our lives, from everyday technologies such as smartphones and socio-technical systems to high-stake decision-making contexts including scarce-resource allocation, loan approval, and automated hiring decisions [27, 66, 67, 73]. While they offer personalized, scalable, and fast decision-making

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services, recent evidence suggests how these algorithmic systems could cause discrimination and harm, and disproportionately impact marginalized individuals [82]. Algorithms can have detrimental short- and long-term effects on individuals, including but not limited to exclusions, loss of opportunity (e.g., job, economic), and stereotyping. Despite increasing harmful impacts, AI/ML continues to become more relevant in every aspect of human life.

As AI/ML[1] applications become more pervasive, AI literacy initiatives have emerged to empower individuals to critically assess and effectively use AI systems [58]. The significance of AI literacy was also echoed in the recent White House executive order for the US workforce and talent development.[2] It is predicted that AI will not replace jobs, however, people with AI knowledge will have a competitive advantage over those who do not [65]. More generally, people with AI knowledge are informed citizens, who are able to interact with AI systems meaningfully and use them responsibly, as opposed to misusing AI. In that vein, a large body of recent works has investigated ways to teach basic ML algorithms (e.g., decision trees, k-NN, supervised ML concepts) and data literacy concepts to K-12 students, middle-schoolers, and non-STEM majors [52, 56, 59]. These studies have shown that it is possible to educate people about ML algorithms, ML's data dependency, and data science concepts, without previous computer science, math, or statistics background. Most of these works do an excellent job of introducing novel ML concepts using educational games and instructional materials. While these initiatives are important, we observed a general lack of interventions in teaching people about the hidden nature of ML biases, and how we can detect and prevent them.

The negative impacts of ML biases (e.g, racial or gender-based discrimination) are both pervasive and often invisibilized [17, 27], resulting in harm to the general public.[3,4] ML biases can creep into ML systems through technical steps in the ML pipeline (e.g., fine-tuning model parameters, model training, and testing), through human biases (e.g., pre-existing biases [33], cognitive biases [35]), problems associated with the dataset(s) (e.g., incomplete or noisy data [40]), or perhaps due to the socio-technical nature of ML biases (e.g., bias penetrating through power and force [17, 48, 82]).

Earlier works have deeply explored how different kinds of biases can manifest themselves in the ML pipeline [94] and in computer systems more broadly [33]. Other large bodies of work primarily emphasized raising ethical awareness about algorithmic harms among various users (e.g., teens [31], and software engineers [4]). Only a few recent works have sought to teach algorithmic bias-related concepts to undergraduate and CS majors using instructional materials (e.g., [28, 37, 75]). We would argue that ML bias knowledge is important for everyone, including technical experts who will be developing ML models and those who could be directly or indirectly impacted by the ML systems and their outcomes [32]. In that regard, this work presents findings from a case study of a transformational game designed to educate post-secondary students about ML biases. The following Research Questions (RQs) guided our work:

RQ1: How might we teach students about ML biases through game mechanics and transformations?
RQ2: How effective was the game in enacting the desired change in the player?

In this paper, we designed a transformational multiplayer game, *"Equality Engine"*, for teaching ML biases and debiasing concepts through a playful immersive learning environment. Informed by

[1]In this study, we do not strictly differentiate between AI and ML bias, and sometimes use these terms interchangeably, as it is common in public discussions. However, we acknowledge that more technically oriented prior work treats them distinctly. For example, Saltz et al. [77] confines ML ethics issues to those related to the technological underpinnings of ML, whereas AI ethics encompasses broader concerns such as job loss and warfare.

[2]https://www.whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/

[3]https://www.cnbc.com/2023/12/28/in-the-job-hiring-process-most-workers-say-they-already-sense-ai.html

[4]https://www.nytimes.com/2016/09/19/us/housing-bias-and-the-roots-of-segregation.html

related works in [16, 61, 94], we prototyped a digital board game consisting of the five stages of ML system development and the five sources of biases associated with each stage. We conducted a mixed-method formative and evaluative playtest study of our game with 12 STEM post-secondary students (6 pairs) who had varying levels of prior ML experience. Both qualitative and quantitative findings showcased that our game has increased participants' self-reported ML bias-debias knowledge. Additionally, participants shared their willingness to use the game in the future due to the knowledge they gained. They found the game easy to use because of the social interaction and immersion the game enabled.

We offer the following concrete contributions in ML literacy and ethics scholarship:

(1) The design of *Equality Engine, a transformational online board game about ML biases and debiasing techniques* as well as the relationship between ML systems and real-world situations and impacts.
(2) *Mixed-method playtest case study findings* showcased the game's effectiveness in introducing players with varying ML expertise to different kinds of ML biases. The game successfully engaged players in meaningful discussions about ML bias and debias techniques, while significantly improving their self-reported knowledge of ML biases.
(3) *Game mechanics* that enable complex ML bias concepts digestible, *facilitating social interactions, collaboration, negotiations*, and opportunities for *critical reflection*. We also offer design implications for future games aimed at ML bias education.

2 Background and Motivation

In this section, we briefly survey the existing literature on AI literacy, followed by research focusing on ML education for technical and non-technical learners. Finally, we review the literature on game-based intervention in ML education. Throughout, we will highlight key areas where this research is making a contribution.

2.1 Trends in AI Literacy

The widespread growth and application of AI in business, industry, science, and healthcare have improved efficiency and enhanced users' personalized experience. AI has become inseparable from everyday life [64, 65]. Similarly, we observe an increasing demand of AI education among citizens worldwide. AI is a broad term that encompasses applications ranging from smartphones and home appliances to more complex machine learning-driven systems, such as ChatGPT and Google. Given the multi-disciplinary and widespread applications of AI, research has sought to define 'AI literacy' [58, 64]. In this paper, we subscribe to Long and Magerko [58]'s definition of AI literacy as *"a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace"* [58, p.298].

To facilitate and support individuals' access to knowledge, and to help them apply, understand, and create ethical and inclusive AI [23], different AI literacy paradigms have emerged. This includes computational literacy (e.g., teaching how to debug code [1]), digital literacy (e.g., teaching how to use computational applications [72]), data literacy (e.g., interventions teaching training data [88]), informational literacy (e.g., education about algorithmic systems [87]), among others. AI literacy studies also placed increasing attention on spreading AI literacy among children, K-12 students, higher education students, and older adults [23, 53, 64]. However, there is a widespread lack of AI literacy, particularly regarding AI ethics, such as bias in AI [58, 65]. Research suggests that AI's influence has not always been positive; oftentimes AI's harmful impacts are dismissed as unavoidable, particularly for marginalized populations who are disproportionately affected. For example, facial classification systems produce biased outcomes against racial minorities [66], and automated hiring applications perpetuate gender-related biases [67, 73].

In the field of computing education, the research community is increasingly emphasizing the importance of preparing the next generation of upcoming leaders to understand the potential impacts of technology systems. Significant efforts have been directed towards integrating ethics education into computer science education [69], mainly through curriculum development [22, 29, 85]. With the rapid advancement of AI and ML techniques and the ethical dilemmas they present, there is a growing focus on enhancing the AI/ML ethical curriculum, especially for practitioners. This includes developing course modules [13, 25], activities [77], assignments [12], and methods for analyzing algorithms [55] to help learners incorporate ethical considerations into technical practices across various topics such as Machine Learning [25, 77], Predictive Algorithms [13], and Natural Language Processing [20].

However, efforts to raise ethical awareness among other audiences, who have low AI technical expertise, nonetheless directly impacted by the outcomes of AI applications in their daily lives, have been limited. For example, in the review by Laupichler et al. [53] on AI literacy in higher and adult education, fewer than one-third of the 30 papers specifically addressed ethical issues. In this paper, we aim to contribute to the research addressing the pressing need to improve AI bias literacy among post-secondary students. This involves fostering future responsible citizens who can use AI in an informed and reliable manner [65], promoting broader participation in auditing the ethical impacts of AI at societal levels [74], and ensuring that the algorithms governing our lives reflect desirable human values, such as fairness. In this paper, we specifically focus on ML bias literacy to contribute to the growing body of AI/ML ethics scholarship.

2.2 HCI Research on ML Education

Teaching machine learning (ML) is challenging, as it builds on foundational knowledge from disciplines such as statistics, algebra, and computer science. Thus, ML and data science knowledge has been primarily accessible to scientists and scholars in STEM-related fields. To make ML more accessible outside of STEM-relate field (e.g., marketing teams, UX designers), tools such as automated machine learning concepts have emerged (e.g., AutoML [44]). However, their adoption has been limited due to the lack of transparency and the challenges associated with merging them with users' existing workflows [92]. Despite such innovations, ML concepts are not easy to consume without relevant prior experience.

A recent line of work focused on broadening technical ML education to a wider population using ML algorithmic literacy and data literacy [52, 56, 59]. Some of these contributions focused on teaching basic ML algorithms to K-12 students and non-STEM majors. For example, Ma et al. [59] and colleagues developed instructional materials for teaching decision trees and k-nearest neighbor algorithms to middle school students (i.e., children aged 12 to 14). Hands-on activities and intuitive examples, such as hand-drawn decision trees and penguin species classifications, facilitated students' learning by enabling them to optimize results on a smaller scale. Another work also developed instructional materials and showed how the use of smilies and games can support non-STEM students (e.g., marketing, law, management) gain important ML concepts and steps in the ML pipeline. Students learned about how to create R scripts and apply predictive models for exploratory data analysis in marketing and finance applications [36].

Data is an essential part of all ML-powered systems. Thus, besides algorithm and ML literacy, data literacy has emerged to broaden the participation of different populations on data. These data literacy studies investigate methods for teaching people about data and engaging them in critical discourses about its implications [14, 88, 89, 100]. On one hand, Calzada Prado and Marzal [14] offered a framework of five core competencies to incorporate data literacy in library instructional design. These five competencies are: (1) understanding data; (2) finding/obtaining data; (3) reading, interpreting, and evaluating data; (4) managing data; and (5) using data. On the other hand, Solyst et al. [88] presented a tabletop card game Data Detectives to teach youth (and families) about training data and showed that high-level game mechanics can support youth participation in critical discourse about algorithmic biases.

The previous paragraphs illustrate a growing emphasis on educating a broader audience, such as K-12 students, children, and youth, about data and ML algorithms. Informed by this large body of work, our work focused on supporting post-secondary students with a range of prior ML experience about biases. ML biases are often subtle and socio-technical in nature, making them difficult to detect and prevent early on. As a consequence, biases can inadvertently infiltrate the ML pipeline. This goal is imperative because post-secondary students who will enter the workforce will become the next generation of primary ML audience, using or even creating the next generation of ML applications, who must have critical knowledge of ML ethics and in-depth awareness of ML biases.

2.3 Games for ML Education

Games are a popular medium for education, covering topics from programming and debugging to computer science and machine learning [1, 47, 63, 88]. These games are also known as serious games that use game mechanics, stories, and game transformation to achieve the intended goal [18]. These games have proven to be beneficial for educating students without computer engineering background about ML. For example, Bae et al. [2] proposed Classy Trash Monster, an educational game for teaching non-majors how to train classification models. Findings showed that students learned about ML basics and the influence of data on ML outcomes. Recent works have also discovered the capacity of games' as a 'soft' medium for teaching ML concepts among non-majors [36].

As ML continues to impact society, games are being developed to teach its technical concepts to wider audiences in various age groups and professions. For example, Priya et al. [70] developed ML-Quest, a game designed to introduce the core concepts of ML such as Supervised Learning, Gradient Descent, and K-nearest neighbor classification algorithm to K-12 students. The results showed that students who played the game outperformed those who learned the same concepts through slides on a subsequent knowledge test. Similarly, Opel et al. [68] developed a simulation game called "Man, Machine!" that introduces how ML works. This game specifically provides teaching materials for educators of diverse backgrounds and students of all school levels, starting around 12 years of age.

While there has been increasing attention on ML ethics education, very few games have been specifically developed to address ML bias and debiasing concepts, to the best of researchers' knowledge. Furthermore, many of the games developed for ML education were software built that created additional barriers such as requiring plugin or game engines installation [2, 47]. Several works have taken a more accessible approach, specifically through cardboard games, to enhance ethical awareness of technology. One closely related work is by Brinkman [11], which uses a card game in the classroom to engage computer science students with the whistle-blowing ethical responsibilities of a software engineer or computer scientist. This approach helps prevent students from merely professing selfless dedication to meet societal expectations and instead, immerses them cognitively in scenarios where personal interests may conflict with the public good. Meanwhile, several successful cardboard games have been developed to raise non-expert users' attention to the technology-generated issues in daily life, especially in raising cyber-security awareness (e.g., [19, 39, 43, 103]). For example, Denning et al. [19] developed a card game to enhance awareness of computer security needs and challenges, aiming to prepare participants as more informed technology builders and consumers. The game also features hacker characters from diverse backgrounds to emphasize that the information technology community is welcoming to people of all backgrounds [43]. Inspired by these works, we were motivated to adopt a similar approach by designing an interactive web-based board game to raise awareness of ML bias. We ensured that the game could be easily run on any browser and be compatible across all platforms, including desktop and mobile, to make it accessible to a wider audience.

3 Game Design

In this section, first, we outline our purpose and goals in designing *Equality Engine*, a transformational game for machine learning bias literacy. We then provide details of the game's rules and components, followed by a sample gameplay interaction.

3.1 Transformational Game Goals

Equality Engine is a transformational game (i.e., a game that changes the player on purpose [18]) that aims at enhancing ML bias literacy of our target audience - post-secondary students - through interactive game mechanics. We followed a structured transformational game design and research process [18, 98]. Our first step was to select our transformational goal, i.e., introduce core machine learning principles that would be helpful in the day-to-day life of a novice audience. We then entered the literature review phase to select those principles. After several rounds of iterative design, we decided the game should:

(1) *Improve awareness of potential ML biases*: participants should be introduced to ML bias concepts (e.g., race, gender, geographical biases), potential sources of harm throughout the ML pipeline, the impact of ML biases on the model behavior and ML system outcomes, and countermeasures that can be taken to mitigate or prevent those biases.
(2) *Highlight the interwoven relationship between ML systems and real-world situations*: participants should learn about the impact of ML applications in everyday life circumstances, the impact of biases on the behavior of ML applications, and current efforts to mitigate ML biases or debiasing techniques.

We conducted multiple rounds of internal playtesting and prototyping, following best practices for research-backed transformational games [98]. The final version of *Equality Engine* is a cooperative, two-player (or multiple pairs) board game, where each player takes on the role of an engineer debugging a machine learning algorithm for hiring by making decisions about the tradeoffs of different biases present in ML algorithms. A playable version of the game, game design materials (i.e., bias and debias cards), and study questionnaires can be found in the GitHub repository. [5]

3.2 Key Game Components

The game was developed using PlayingCards Virtual Tabletop to enable a remote gameplay experience. The game board, bias, and debias card decks are the key components of the game. [6]

3.2.1 Game board. The proposed game scenario is based on the development of a fictional ML system that would be deployed for hiring post-secondary students. Informed by earlier work [16, 61, 94], the game board presents the five stages of ML system development and the five sources of biases associated with each stage.

The game board was designed to be easily understandable among students with varying levels of ML knowledge and background (see Fig. 1). To achieve this, at first, we simplified the ML pipeline and sources of biases as they travel through the ML pipeline. Additionally, we selected examples of

[5]https://github.com/wang0q/EquityEngine
[6]https://playingcards.io/

each type of bias, scaffolding the understanding of abstract bias definitions (see Section 3.2.2 for examples in the bias card deck). Our selection of biases is grounded in the framework by Suresh and Guttag [93], who identified seven types of bias throughout the machine learning life cycle: historical, representation, measurement, learning, aggregation, evaluation, and deployment bias. Our research team collaboratively selected historical, representation, measurement, and deployment biases for their accessibility and ease of understanding, even without prior ML knowledge. We also added a fifth bias category called modeling bias. "Modeling bias", which is composed of learning, aggregation, and evaluation biases. This consolidation was done for two main reasons: (1) these three biases occur during the ML model-building stage; (2) discussing these stages separately could be too complex for students with varying ML experience. For instance, learning bias is defined as biases that arise when *"modeling choices amplify performance disparities across different examples in the data."*[93] Explaining this with multiple models and datasets would likely overwhelm the students. Consequently, a total of five biases are presented on the game board.

Secondly, after bias selection, we selected a scenario closely related to our target audience; we also visualized abstract ML concepts in the chosen scenario. Using visualization and real-world applications or contexts has been identified as an effective pedagogical approach to reduce learning barriers for non-major students [90]. We selected the hiring scenario because post-secondary students can relate to it more easily, as they will soon enter the workforce, and a lack of awareness of the sources of biases in hiring could leave them unprepared. Thus, obtaining insight into ML systems, such as automated hiring processes, is critical for them. We designed the game in response to the urgent need to recognize and counteract persistent biases against BIPOC students in hiring processes, as highlighted in recent research [73]. [7] Additionally, by emphasizing making moves voluntarily, like 'helping', 'discovery', 'removing', etc. (refer to Section 3.3 for more details), the game actions encourages all participants to proactively address these issues, whether they are BIPOC students seeking to empower themselves or non-BIPOC students aiming to promote camaraderie in our diverse society. This game scenario, therefore, not only educates but also fosters critical awareness among students about the broader implications and challenges of ML deployments in real-world settings.

3.2.2 Bias Card Deck. The bias deck showcases five sources of bias: historical bias, representation bias, measurement bias, modeling bias, and deployment bias. These biases are also presented on the game board (see Fig. 2). These five bias types are sourced from the framework proposed by Suresh and Guttag [94]. We selected four examples, for each of the five types of bias, resulting in twenty examples of bias. The research team iteratively revised the examples and language, ensuring they were relatable and easily understandable to our target audience. See Table 2 in the Appendix for definition and example cards with sources. To aid identification, each category of the bias has been placed and color-coded in the title. Example cards from each bias category, along with the front and back sides (which are the same for all cards) are presented in Figure 2.

Specifically, most cards contain an example of the source of bias in the hiring context, translated from [73]. For example, we have reinterpreted the observation by Raghavan et al. [73] that certain algorithms, originally designed using general assessments from broad datasets, often fail to perform adequately in specialized job contexts within a specific company. We have categorized this under 'Deployment Bias', terming it 'Problematic Generalization.' Additionally, we have enriched our analysis with examples from other scholarly sources. Notably, in discussing 'Modeling Bias,' we reference the well-documented case of Amazon's AI, which, despite being trained on gender-neutral data, exhibited a discernible bias against women, as highlighted by Kodiyan [51].

[7] BIPOC stands for Black, Indigenous, and people of color. It acknowledges that people of color face varying types of discrimination and prejudice.

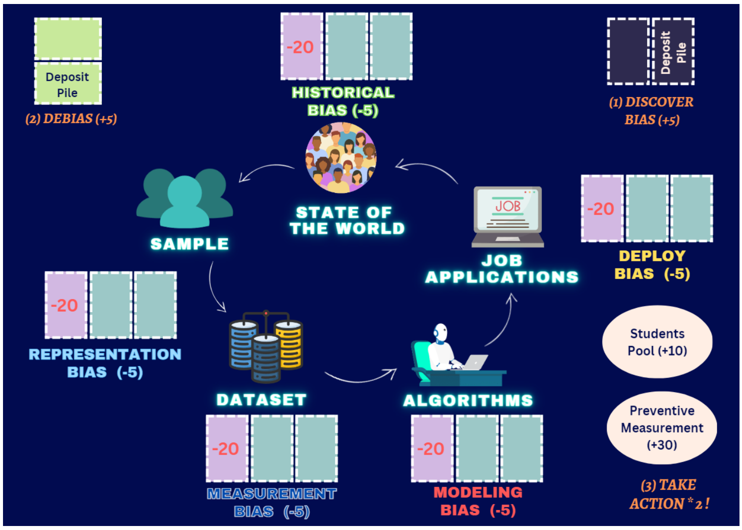

Fig. 1. Game board for *Equality Engine* ML bias literacy game.

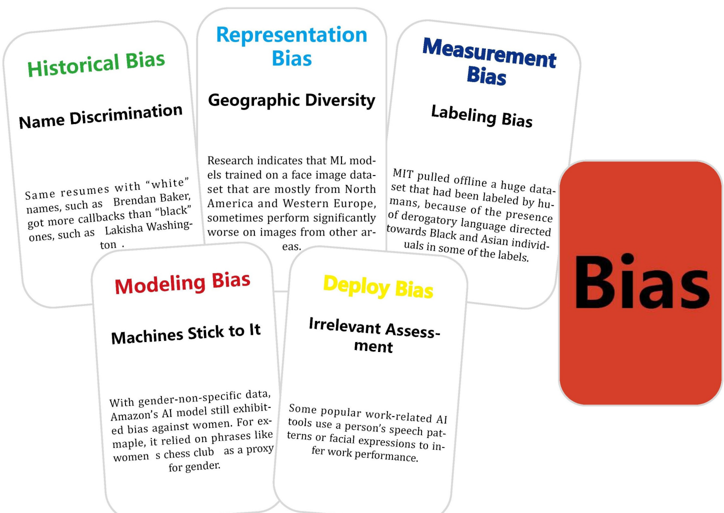

Fig. 2. (on the left) five examples of each bias card category, and (on the right) back side of bias cards.

To ensure a balanced representation of bias categories, which enhances both the educational aspect of our study and the exposure of our audience to a variety of biases, we have included more general examples. These instances, while not directly addressing hiring, are important in the discourse on how biases fundamentally affect minority groups. For instance, in exploring 'Representation Bias,' we discuss 'Geographic Diversity', highlighting the issue with many ML models trained on face image datasets predominantly comprising individuals of European descent. This lack of diversity often leads to worse performance when processing images of people from other ethnic backgrounds, as reported by Serna et al. [81]. Furthermore, we have incorporated examples from the media that underscore the impact of bias in the public sector. A notable instance

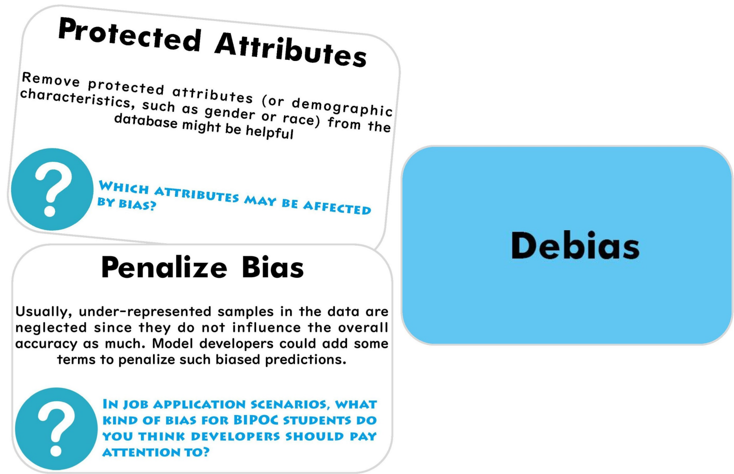

Fig. 3. (on the left) two examples of debias cards, and (on the right) back side of debias cards.

is the 2020 removal of a large dataset by MIT due to the presence of derogatory language targeting Black and Asian individuals in some of the labels, an event reported by The Register. [8]

Additionally, we have endeavored to mitigate the cognitive load associated with assimilating new information from the examples provided for the players. To achieve this, we have implemented two strategies: (1) we simplified the technical aspects of research citations, ensuring the preservation of their core concepts. For instance, in the card "Facial Recognition Bias" (refer to Appendix 2 for full details), we rephrase the technical statement that *"disparities in error rates across gender and racial lines"* [73]. We clarify this by providing a specific example: if the training dataset predominantly consists of photos of white men, the facial recognition system will likely underperform when identifying the faces of women and people of color. And (2) we introduce succinct subtitles to encapsulate the essence of each example vividly. For instance, in the Amazon AI case, we employed the subtitle 'Machines Stick to It' to concisely communicate the idea that biases can inadvertently become embedded in machine learning models, leading to unintended outcomes. This is exemplified by Amazon's algorithm, which learned to unfavorably evaluate resumes containing the word "women's", as discussed by Kodiyan [51]. Such a concise representation underscores the urgency of addressing biases in machine learning systems. These adaptations and subtitles were collaboratively refined by our research team through multiple rounds of iterations and further validated for clarity and comprehension through a pilot study. Our research team is composed of computer scientists, PhD students, and research assistants having a broad range of expertise in game design, Psychology, CS, ML, and Engineering. Collectively, we bring over a decade of design and research experience to promote equity and inclusion in STEM, computing, and beyond.

3.2.3 Debias Card Deck. The debias deck presents counter measurements for ML bias, accompanied by thought-provoking questions. Each card presents a debias method, including a textual description, and a title summarizing the method (see Fig. 3). We also included a question to prompt players to contemplate the practical application of these methods in real-world scenarios and engage players in collaborative discussion, which will be further discussed in section 3.3. See Table 3, in the Appendix, for definitions and example cards together with source information.

[8] https://www.theregister.com/2020/07/01/mit dataset removed/

We introduced six debiasing approaches. Our selection leaned towards overarching principles, such as advocating for interdisciplinary collaboration and the critical examination of biases within development teams. Additionally, we included more intuitive examples, even if they may not represent the most recommended debiasing practices or they lack the complexity and sophistication needed for real-world application. For example, in the 'Protected Attributes' debias card, we suggest the removal of certain sensitive attributes, like demographic characteristics, from datasets. Another debiasing technique called 'Penalize Bias', requires model developers to incorporate certain terms into algorithms, to actively penalize biased predictions against under-represented populations. These methods, while simplified, provide an accessible introduction to the debiasing concepts in machine learning. Similar to bias card game content, debias card contents were collaboratively refined by our research team through multiple rounds of iterations. Moreover, these card contents were further verified for clarity and comprehension through a pilot study.

3.3 Game Mechanics and Play Dynamics

The game mechanics are structured into three phases: Discover Bias, Debias, and Take Actions. A complete pictorial workflow of the game can be found in the Appendix, in Figure 6. During a standard game session, two players engage in three rounds of play. In each round, each player sequentially progresses through these three phases. At the end of the game, the team's total points are calculated, and a leaderboard on the game board records the top three historical scores. All the game phases, action options, and points effects are visualized on the game board for players' reference (see Figure 1).

It should be noted that the game could support multiple pairs, with each team competing to achieve higher scores. However, considering the constraints in participant recruitment and time allocation, along with the intention to avoid imposing undue competitive pressure on participants, the experimental design was adjusted. We limited the game to two-player interactions per game and introduced a leaderboard to serve as a motivational element rather than a direct competitive feature. Further details on the three game phases are given below.

3.3.1 Discover Bias. One player draws a bias card from the pile, reads its content privately, and explains the bias to their teammate using the provided text description and subtitle. However, they are not allowed to mention the bias category, which is the card's title. The teammate can ask for additional explanations or do a re-read to identify the bias category and provide an answer. If the teammate correctly identifies the category, the team earns five points. Regardless of the correctness, the bias card is placed in the corresponding card holder on the game board.
Examples: (G003)

```
P006 (read the bias card content): The Bureau of Indian education has repeatedly
neglected warnings that it does not provide a quality education for 46,000 native
students.
P005: It is historical bias.
P006: Yes, that's correct.
```

3.3.2 Debias. One player draws a debias card, reads the example and questions aloud, and collaborates with their teammates to answer the questions. If the team formulates a suitable answer and explains it effectively, the player retains the debias card, and the team earns five points. Participants were encouraged to go beyond the debias card content while remaining under the topic presented on the card. They were asked to come up with innovative responses for specific questions and were allowed to use in-game learnings or previous examples for these responses. Participants were asked about technical approaches, human involvement, and related strategies for bias removal (see

Table 3, in the Appendix, for examples). When participants gave slightly diverging or repetitive debiasing strategies, the researcher subtly prompted them (e.g., "Can you give another example?") to encourage their creative thought processes. These responses were meant to gauge learning and engage players in further discussions. Thus, players were given ample opportunity to do so. Hence there were no right or wrong answers in this phase. The researcher evaluated the responses based on uniqueness, creativity, relevance, and positive impact, and once satisfied, participants were rewarded points for it.

Examples: (G006)

```
P012: Do you have any idea?
P011: I think they can like, create their own survey. Is that is that the one you . . .
let me think about it for a while
P012: Oh, one thing that I think can also be done is that the teams can get a
third-person perspective from someone else like they can invite someone else and
discuss whatever they have come up with that person. and then try get the opinion
of that person, whether or not it is biased, and to what extent it is biased.
If it is done, 3 or 4 times like with 3 or 4 different people, then they would get a
good idea about whether or not they have bias.
```

3.3.3 *Take Actions.* Players can choose two actions from the following four options:

(1) Help a student secure a job by moving a "BIPOC student" marble to the "Student Pool" area on the game board, resulting in a gain of ten points.
(2) Remove a bias card from the game board, which does not grant additional points but prevents future points loss. This action is crucial because:
   (a) At the end of each round, if there are more than two discovered biases in one category, indicated by two bias cards in the corresponding area on the game board, it leads to a loss of ten points.
   (b) At the end of the game, each bias card remaining on the game board results in a five-point loss.
(3) Provide a debias card to a team member, which does not yield immediate points but may prove useful for action type (4).
(4) Create a preventive measurement by exchanging three debias cards in hand for a "Preventive Measurement" and putting it in the corresponding area on the game board, resulting in a gain of 30 points.

Examples: (G06)

```
P011: Okay, I will give you one debias card.
P012: And along with that. Can we like, remove one bias. Because they are just going
to keep adding up.
P011: Okay. But at the same time one bias card is minus 5 points. Well, if you have a
students plus 10, so. . .
P012: That is at the end of the entire game. But at the end of one round one. It like
it will cost us 10 points.
P011: Yeah, but don't. We just have like one card. Now for each for each turn, like
If you put students into the student pool, we get plus 10 points for whole round
P011: Right, and if you move the card, it's plus 5 by reducing the minus 5 points
at the end of the round for each term. So
P012: That makes sense. Yeah.
```

In summary, these game mechanics are designed to create an engaging and strategic gameplay experience, where players must contemplate upon bias categories, engage in meaningful discussions to address debiasing challenges, and work together collaboratively and thoughtfully during the Take Actions phase to maximize the team's score and overall success.

4 Methods

We conducted a mixed-methods playtest focus group study to evaluate *Equality Engine's* effectiveness in introducing ML bias concepts and to solicit formative feedback on the game.

4.1 Participant Recruitment

The purpose of this study is to playtest a transformational game designed to introduce machine learning biases to post-secondary students and explore how these biases impact them in the job hiring context. To be eligible for participation in this study, participants must meet the following two criteria: (1) they must be 18+ years of age, and (2) enrolled in a college or university, or be a recent graduate. During participant recruitment, study flyers were posted at popular locations throughout the first author's university, including study spaces and cultural centers around the campus. The study was also advertised online on university-specific websites and social media. The flyers and online advertisements directed prospective participants to a survey requesting information such as age, gender, race/ethnicity, and prior ML experience.

During the playtest sessions, participants were organized into groups of two. Since the game content involves extensive discussions on ML ethics-related hiring biases, participants were paired based on their race to ensure comfort during discussions about potentially sensitive racial experiences (e.g., [76, 96, 97]). As such, this intentional grouping strategy, along with scheduling conflicts between participants themselves, resulted in a limited number of final players selected to participate in the study, totaling 14 students. To schedule these playtests, we sent emails with learning materials so the participants could familiarize themselves with the mechanics of the online website platform before the study. Participants were granted a $15 award for their contribution.

Studies were conducted by two researchers through virtual Zoom conferencing. One researcher was responsible for facilitating gameplay and leading discussion, while the other recorded notes. To begin the study session, participants were given a brief introduction to the five types of biases essential to gameplay. Afterward, the participants could ask any clarifying questions before starting to play. All studies have been reviewed and approved by the Northeastern University Institutional Review Board (IRB).

4.2 Test Plan: Data Collection and Data Analysis

In total, we collected over 12+ hours of gameplay data. We recorded both audio and video gameplay sessions. In total, 14 players (7 groups, two players per group) participated in the playtest. However, one group did not submit the post-gameplay survey, so our analysis included 12 responses. All participants were undergraduate or graduate students (age: range 18-28; mean = 22; SD = 2.82). Among 12 participants, four participants self-identified as men (33.33%), and eight participants identified as women (66.67%). Racially, one participant self-identified as Indian (8.3%), two participants East-Asian (16.67%) and the rest of the participants identified themselves as Asian (75%). Our participants included all STEM majors. STEM participants' majors included CS, data science, health science, and Pharmaceutical Sciences. During the focus group sessions, four participants indicated they had no previous experience in ML or had taken related courses, while eight participants had ML experience in general through relevant coursework.

We applied a mixed-method study design for the playtest to solicit formative and evaluative feedback about the game's educational impact as well as the overall gameplay experience. The playtest was composed of three sessions.

(1) First, in the pre-gameplay session, participants completed a pre-gameplay survey that measured their self-reported prior ML knowledge, self-efficacy towards ML [99], and ML-related anxiety [79]. ML knowledge composite measure (i.e., consisted of four survey items [5]) was constructed based on the game design. Self-efficacy and anxiety towards ML were measured using slightly modified questionnaires, adapted from [79, 99] to fit our study context. These scales were important for answering specific RQs. For instance, we were interested in students' ML bias knowledge improvement. We were also interested in measuring ML self-efficacy and ML anxiety because students perceived high self-efficacy and low anxiety levels are important factors in their learning.
(2) Second, in the gameplay session, participants played the game in pairs. All conversations were both audio and video recorded. Playtesting was followed by responding to several open-ended interview questions that asked participants about their overall experience, i.e., aspects of the game they liked, and didn't like, including suggestions for game improvement.
(3) Lastly, participants finished the post-gameplay survey. The post-gameplay survey asked participants to self-assess their change in ML bias knowledge and self-efficacy. Participants self-reported change in ML bias knowledge and self-efficacy was measured using the same questionnaire in the pre-post test. In addition to facilitating students' ML bias learning, we were also deliberate in understanding whether the students found the game useful or not, and whether the learning experience was enjoyable, therefore we included the game's utility and enjoyment measures. Precisely, in the post-gameplay survey, we also asked participants to evaluate the game's usefulness and game enjoyment using exact questionnaires from [43] and [34] studies, respectively.

All the composite variables introduced above were measured on a scale of 7. Each measure was created by taking the sum of all ratings/responses. [9] Survey measures were collected using the Google Forms platform. All quantitative responses were exported into a .csv file, then pre-processed, and analyzed in R. The reliability of all non-categorical survey measures was tested for internal consistency at an acceptable level (i.e., $0.7 \geq$ Cronbach's alpha ≤ 0.9). [10] The complete survey instrument can be found in the GitHub repository.

To curate our qualitative data, we first exported transcripts (.vtt files) from Zoom and cleaned them. We then separated this data into four categories: 1) discovering bias question responses, 2) debias discussions, 3) all remaining gameplay conversations, and 4) post-gameplay interview responses. We used a mix of deductive and inductive thematic qualitative analysis [7, 10]. Thematic analysis is flexible, one of the most widely used foundational methods for qualitative data analysis in HCI and beyond [6].

The discover bias question responses were coded deductively. These responses were coded as correct = 1 or incorrect = 0 directly from the script, as the game moderator assigned points based on the accuracy of those responses during the playtest. For the remaining three categories of qualitative data (i.e., debias discussions, gameplay conversations, and post-gameplay interviews), we used inductive thematic analysis [7]. Themes were actively generated as part of an active and creative process by the researchers through their interpretative engagement with the data [8, 10]. Our team collected data and analyzed them in parallel, as the study progressed. All researchers met regularly, shared each other's analysis, engaged in discussion about similarities, and worked

[9] https://www.statisticssolutions.com/composite-scores/
[10] https://www.statisticssolutions.com/cronbachs-alpha/

together in case of any novel or surprising findings or incongruence. The themes were reported only after reaching a consensus.

Discovering bias responses were one sentence long at most, however, the debias responses were longer and consisted of multiple paragraphs (i.e., each paragraph contained three or more sentences). Debias responses were used in qualitative analysis to report and capture commonly recurring themes, patterned responses, shared meaning, shared topics, and all other aspects participants considered valuable – to tell a coherent narrative from data. Lastly, the playtest conversations and the post-playtest interview discussions were analyzed to capture interesting and useful themes. We stopped conducting further studies as we began to observe repetitive themes [9].

5 Results

5.1 RQ1: Participants Perception and Understanding of ML Bias-Debiasing Techniques

To gauge the effectiveness of this game in machine learning bias literacy, we facilitated a post-gameplay discussion that prompted participants to reflect on their level of learning, game enjoyment, and overall experience. These transcribed post-gameplay discussions and gameplay transcripts revealed two emergent themes about students' perceptions and understanding of ML biases. Specifically, these themes are: (1) participants felt that the game was both stimulating and educational; (2) the game enabled them to carefully consider the role of ML in the context of the state of the world.

5.1.1 Learning About Various ML Biases. In the Discover Bias phase of the game, students collaborated to identify the type of bias presented on the bias card. To gauge learning ability through gameplay, we quantified the number of correct [11] and incorrect [12] biases across each bias type and group session (See Figure 4). Our results indicated that students were most successful at distinguishing historical biases and least successful at determining representation biases. Precisely, 37.5% of all correct bias inferences were historical bias whereas 6.25% of all correct bias inferences were representation bias. Conversely, students were most likely to misidentify representation bias and least likely to misidentify modeling bias, i.e., 40% of all incorrect bias inferences were representation bias whilst 5% of all incorrect bias inferences were modeling bias. Unsurprisingly, the group with players that had no prior experience with machine learning had the least number of correct bias inferences and the most number of incorrect bias inferences. Perhaps, this indicates that students with some level of prior ML knowledge (either through class or out-of-class informal training) were able to understand the concepts presented in the game more effectively or due to some high-level knowledge and understanding about ML.

Binary outcome, i.e., correct/incorrect based bias-inference mechanics and associated score (i.e., +5 points) were an interesting choice from the game design perspective because it required students to pay attention and a certain level of memorization. However, the natural complexities and relatedness in bias scenario descriptions might make it difficult to measure the game's teaching ability using correct/incorrect scores alone. For instance, students' responses highlight an important aspect of discussions around biases, which often involve varying subtleties and nuances depending on the context of the scenario presented in the cards. These nuances in the biases make them challenging to fully grasp, while they raise inquisitiveness, at the same time they also cause slight confusion. For example, students (G007) mistakenly guessed representation bias (e.g., "It is impossible to determine how the current employees compare to the entire pool of candidates because there is no data available on the performance of those who were not hired") as historical bias. While students did a good job of determining modeling bias (see Fig. 1), they also made more mistakes in guessing them correctly, often confusing them with other types of biases (e.g.,

[11]Correct bias inferences were defined as when students successfully identified the type of bias.

[12]Incorrect bias inferences were defined as when students misidentified the type of bias.

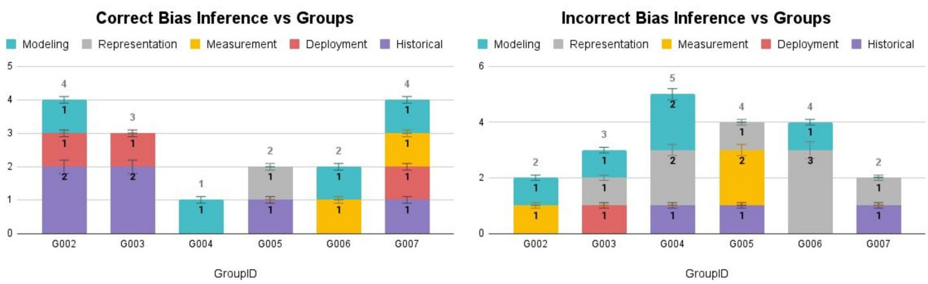

Fig. 4. Correct and incorrect bias inference measures across all groups, in all three rounds of gameplay.

representation bias – G004, G006; measurement bias – G005). These examples demonstrate the challenge of capturing nuanced subtleties in various ML biases into scrutable textual descriptions that are easy to consume. Therefore, the game, while effective in introducing these concepts, may leave room for participants to seek more in-depth explanations and clearer guidelines on recognizing and addressing bias, especially in complex or less obvious scenarios. This feedback could be an opportunity for refining the game and the accompanying educational materials to provide participants with a more comprehensive understanding of bias in various contexts.

The postgame play discussion revealed that when participants were asked to describe their gameplay experience and how, if at all, it was conducive to their learning, both students with and without ML experience shared that they were able to glean new concepts from the game, though to varying degrees. One student, who is a computer science major, stated *"I've participated in many guest lectures about computer science ethics for my CS classes, but I've never covered deployment bias. So this is a new concept that I learned today."* Another student, who is a STEM student but does not study computer science, said *"This game does help understand a lot of things, but I wish I had a better understanding of why my bias inferences were wrong."* This sentiment could stem from the nuanced and somewhat subjective nature of bias categories. The fact that biases can be context-dependent, not always straightforward to identify, and sometimes they are highly interrelated might have contributed to the student's desire for more clarity. This being said, learning about various biases was critical to the game debiasing phase, particularly to understand how those ML biases can potentially be removed either using technical or human interrogations.

5.1.2 Meaningful ML Debiasing Conversations. In the Debias phase of the game, students engaged in a collaborative discussion to discern possible methods of removing bias presented in the debias cards using ad-hoc strategies. The moderator awarded the team points according to the conclusions from their discussion. All groups, regardless of their ML background or performance in the Discover Bias phase, engaged in thoughtful discussion and were awarded points for their conclusions. For example, when prompted to address the question *"What experts can you seek to help address bias?"* in the Consult Experts scenario, two groups said they would consult sociologists who are more knowledgeable about implicit biases in the real world than engineers or computer scientists alone — underscoring how players started to consociate machine learning biases with the systemic difficulties faced by disadvantaged communities in reality.

Importantly, the Debias phase tasked players to consider potential solutions. When prompted to discuss ways to identify biases while working within a project development team, one group suggested that all members of the team should use online psychological tests to evaluate potential preconceived implicit biases before developing their software. They agreed that being self-aware was *"a good place to start"* because it both encouraged developers to be cognizant of their prejudices and served as a preventative measure from the inadvertent inclusion of such biases from the onset. Crucially, the Debias phase compelled players to recognize that developers have a profound responsibility to actively address and mitigate biases to create fair machine learning systems that produce equitable outcomes. As the players were computer science students, the game motivated them to employ their solutions in their own or future work practices. Processes like removing or swapping protected attributes or demographic features (e.g., age, religion, location, language) were discussed. In addition, players shared intricate technical debiasing strategies such as tuning different model parameters and performance metrics and observing its impact on outliers and imbalanced datasets, e.g., *"Maybe [developers] can change the evaluation metrics of their algorithm and tune [...] like they can try different combinations. … see what is the most efficient one."* - G005. In addition to technical debiasing measures, students also discussed the importance of human verification through different strategies such as making the algorithm publicly available for verification, utilizing private auditing by non-governmental bodies, reviewing models by different teammates, and organizing diverse panels consisting of people with different domain expertise (e.g., G002 would consult a sociologist, G004 would consult ML experts) among others. For instance, to create an effective human review panel for ML bias eradication, students shared, *"I think that the review panel should be a diverse panel with the... If we are talking about individuals and like humans, then individuals from different backgrounds rather than having individuals from similar backgrounds on the same panel, because a different background will promote a different viewpoint and that way we can have a more holistic review."* - G006

While players without prior ML experience struggled to answer correctly in the Discover Bias phase, their performance in the Debias phase was comparable to their counterparts with experience. When prompted to answer the same question as above, the group where both players lacked prior knowledge answered that it would be beneficial for members of the development team to engage in conversation with each other, to identify latent biased presuppositions before building the machine learning model. Similar to the other group of computer science students, their goal was to prevent these biases from infiltrating the development process right from the beginning. For example, students discussed bias-penalizing approaches in the job application scenario, they shared, *"I guess [developers] should pay attention to like the name on the resume. The race at their end, and perhaps some of the background, such as the school they want to, or probably the area they grew up in because some race might grow up in another area than other students do."* - G006. Their answers reflected their thought process: how they recognized that it was the ethical duty of developers to ensure that the systems they created were not only technically proficient but also ethically sound, capable of fostering fairness and equity in their applications.

In the post-gameplay discussion, most players agreed that Debias discussion was the most compelling aspect of the game. One participant described that the Debias discussion allowed them to *"actually think about their ideas and not just have one correct answer."* Another participant, who studied computer science in their college courses, mentioned, *"I've heard that this [machine learning bias] is an issue, but not really much of— how can we fix this issue? So it was interesting to think about it myself."* Thus, these comments elucidated how the game successfully prompted participants to consolidate and apply the information they learned while playing.

5.1.3 Collaborative Strategizing and Negotiations. Players actively exercised collaboration and negotiation throughout gameplay, showcasing their collective exploration of ideas, mutual decision-making, and the exchange of perspectives. Most prevalently, this behavior was exhibited in the Take Action phase, where players are required to consider how their decisions (whether to do two of the following actions— give a Debias card, help a BIPOC student get a job, or remove a Bias card from the board) will affect their overall score.

For example, two players debated either removing a bias card or giving a student a job. After one player suggested removing a bias card from the board during their turn, *"Can we remove one bias card from the board, because they are just going to keep adding up"* - G006, their teammate interjected: *"But at the same time one bias card is minus 5 points. If you help a student, it's plus 10 points."* In another example, one player asked the other if he had any advice during the Take Action phase: *"I'd like to ask my teammate what he thinks would be a good action"* - G006. To which, her teammate convinced her to collect Debias cards for an additional 30 points during the final round of the game: *"Why don't we take the Debias card this time?"*. This interaction highlights the participants' willingness to collaborate, and also shows their deeper level of engagement and understanding of game mechanics. Game scores indeed varied depending on participants' collaborative actions, particularly in the "Take Action" phase, as this step consists of different choices and actions, yielding different outcomes. Interestingly, these collaborative interactions helped participants score the highest compared to those who collaborated or leveraged these strategies less. For example, G003 and G006, who applied long-term strategic planning such as "preventative measurement", scored 80 and 90, earning the second and first spot on the leaderboard, compared to groups such as G002, who didn't use such collaborative strategizing. Instead, G002 leaned toward applying straightforward actions (e.g., removing a bias card, giving a job to the student), resulting in a low score at the end of the game. Regardless of crude quantitative scores, these game interactions also showcased participants' interest in trying out different game strategies and a deep engagement with the game mechanics. Again, the goal of the game score and a position in the leaderboard was to maintain a level of competitiveness and seriousness in the play, not to evaluate them.

Ultimately, the resolution of the debate was representative of their strategic thinking and their ability to consider other broad factors within the game's context— taking into account evolving aspects of the game such as the number of bias cards that remained on the board, and the remaining rounds of gameplay before completion. These examples underscored the depth of strategic thought and collaborative decision-making involved in the game. Players not only weighed the short-term benefits but also considered the potential long-term consequences, showcasing a nuanced understanding of the game mechanics and an ability to navigate scenarios collaboratively.

5.1.4 Qualitative Findings Summary. The game was carefully crafted with easy-to-use game mechanics, content, and descriptions to provide a seamless ML bias-debias learning experience. In that regard, the game starts with a discover bias phase that introduces students with different ML biases using relatable examples (e.g., job hiring, resume). In the next phase, the students applied various debiasing techniques by going beyond the debias card contents, utilizing both technical and human ethical responsibilities about ML bias removal, and engaging in thoughtful and collaborative discussions. The last phase of the game tied these two phases together and required the application of learnings from earlier phases and some planned negotiations to score high and secure a position on the leaderboard. In that way, the game introduced healthy competition and an opportunity for learning new concepts, in a highly collaborative manner.

5.2 RQ2: Measuring Game's Effectiveness, Evaluation, and Enjoyment

5.2.1 Quantitative Exploratory Data Analysis. We measured pre-post change in participants self-reported 1) ML bias knowledge and 2) ML bias self-efficacy. ML bias knowledge consisted of four items focusing on the presence of biases in the ML pipeline, the harm they can cause, and ways to address those biases. These statements were measured on a 7-point Likert scale (1 = never heard before, 7 = very familiar).

Table 1. Exploratory data analysis for pre-post test measures.

Variables	Center	Spread	Shapiro-normality test
ML bias knowledge (pre)	Median = 16.5	IQR = 5.5	p-value = 0.18 (normal)
ML bias knowledge (post)	Median = 24	IQR = 0.5	p-value = 0.05 (skewed)

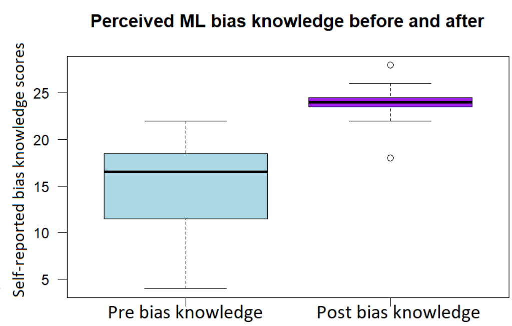

Fig. 5. Participants self-reported pre and post-bias knowledge improvement.

ML bias self-efficacy measured their confidence in learning about ML biases and debiasing techniques through play using a 7-point Likert scale (1=strongly disagree, 7 = strongly agree). Two items in the bias self-efficacy used reverse coding. All pre-post composite measures were computed by summating all items' responses. All pre-post bias measures had four survey items with the 7 highest possible ratings for each item, resulting in the highest possible value of 28. However, the post-self-efficacy measurement included three survey items, giving the highest possible score of 21. Thus, due to the unequal number of items in the pre-post items, the pre-post test was not conducted for the self-efficacy measure. A Shapiro-Wilk normality test was performed on pre-and-post ML bias knowledge. ML bias knowledge (pre) measure had normal distribution, while the ML bias knowledge (post) measure was skewed, so we chose appropriate tests fit for the data, see Table 1. We used correlation and ordinary linear regression to analyze causal relationships. Participants' age, gender, and prior ML experience were used in all regression analyses. In all regression results, SE indicates standard error.

5.2.2 Participants Self-reported Game's Utility for Increasing ML Bias Awareness. Our quantitative findings were in line with qualitative themes. As seen in Table 1, that post bias knowledge score had a skewed distribution (See Fig. 5 for data distribution), thus, a non-parametric Wilcoxon signed rank test with Bonferroni Correction was used. The Wilcoxon signed rank test indicated participants' self-reported knowledge about ML biases significantly improved *(V= 0, p-value = 0.002)* after playing the game with a large effect size d = 1.96. Please note V = 0 here, since the self-reported post-ML bias knowledge is higher than the pre-ML bias knowledge, yielding negative ranks, leading V = 0.

One of the main goals of this game was to raise participants' awareness about ML biases. Therefore, during the post-gameplay session, we measured games' utility or usefulness measure, which consists of three variables: perceived ease of use (PEOU), perceived usefulness (PU), and intention to use (ITU). These scales were informed by prior literature in [43]. These variables were assessed on a 7-point Likert scale (1=strongly disagree, 7=strongly agree). All variables were normally distributed, thus we calculated the mean and SD. Our findings suggest while it took some time for players to learn the game steps, however, at the end of the gameplay, the game, including the game mechanics was easy to understand (PEOU: mean = 10.25, SD = 1.6). Participants also found the game to be useful (PU: mean = 18.25; SD = 1.76), when they reflected on the prevalence of ML. Participants also elicited how they are going to apply their game learning in future professional work practices (ITU: mean = 10.33, SD = 2.34) as they hope to work on various ML applications.

5.2.3 Game Enjoyment. Participants' game enjoyment was measured using the scale in Fu et al. [34] study. Specifically, we measured participants' concentration, immersion, social interaction, and knowledge improvement on a 7-point Likert scale (1=strongly disagree, 7=strongly agree). We measured these scales because they closely presented game mechanics and features in the game. We calculated mean and SD due to normal distribution of these measures.

Participants indicated that the game was immersive (immersion: mean = 36.41; SD = 6.51) such that they almost forgot about time (concentration: mean = 33.25; SD = 4.7) as they reached the final round; some groups even expressed that they wished they could play more rounds to see all the biases. The game also enhanced players' cooperation (social interaction: mean = 41.0; SD = 5.2) and improved their ML bias learning (knowledge improvement: mean = 24.5; SD = 2.06).

We created ordinary linear regression models to discover the effect of game enjoyment measures (i.e., concentration, immersion, social interaction, and knowledge improvement) that predicted the game's utility (i.e., PEOU, PU, ITU) discussed in section 5.2.2 and tested their correlations. The game's perceived ease of use, i.e., PEOU was significantly predicted by the game's ability to engage participants in social interactions *(R^2 = 0.448, F (1, 10) = 8.137, p-value: 0.017)*, SE = 1.291, in the form of collaboration, negotiation, and strategic decision making. PEOU was also significantly predicted by game immersion *(R^2 = 0.385, F (1, 10) = 6.262, p-value: 0.031)*, SE = 1.364. Participants' plan to apply the game in the future was significantly predicted by their perceived knowledge increase *(R^2 = 0.381, F (1, 10) = 6.179, p-value: 0.032)*, SE = 1.936. These results suggest that our game enabled a meaningful, yet playful and enjoyable learning experience about ML biases. Participants' demographics were used in all regression analyses, however, no effect was observed.

5.2.4 Quantitative Findings Summary. Our quantitative findings reveal that students' self-reported knowledge about ML biases significantly improved after playing the game. The game was perceived as useful, easy to use, and enjoyable. Participants perceived the game as easy to use because of the social aspect of the game and immersion. They would also use the game in the future because of the knowledge gained from the game. Although our quantitative findings support the overall positive tone observed in our qualitative data, the slightly negative tone present in the qualitative findings essentially disappeared in the quantitative results. We will discuss this in section 6.2.2.

6 Discussion and Design Implications

Our findings provide insights on how to design enjoyable and easily accessible games for enhancing ML bias awareness. With that evidence, this paper delineates several implications and proposes a future research agenda to design games that enhance ML bias knowledge. This section will first discuss significant game design elements identified through our research — namely, *exposure*, *collaboration*, and *critical reflection* — and examine their role in fostering an effective learning experience. Next, we will propose design implications for future research, addressing the potential and necessary precautions for adapting to a wider audience. We will also explore alternatives for addressing outcome measurement challenges and share our experiences in balancing educational objectives with gamification features in game design. Lastly, we will reflect on study limitations.

6.1 Game Design Elements

Similar card games designed to enhance cybersecurity awareness among non-expert user groups have successfully incorporated features such as fun, accessibility, aesthetic appeal, adaptability, collaboration, and real-life representation [39, 43]. In this context, we discuss three design elements that we find most significant for our ML bias literacy game, tailored specifically to our post-secondary user group, resonating with previous research on computational ethics education.

6.1.1 Exposure. ML bias literacy is contingent upon individuals being exposed to the existence of biases. Prior research indicates that general awareness of the algorithms' shaping of everyday experiences, such as personalized search results, recommendations, and social media curation, is surprisingly low [26, 71]. Moreover, people often fail to recognize systematic biases deeply ingrained in daily life, despite these biases significantly affecting their communities. A notable example is the study by Solyst et al. [89], which revealed that while Black girls noticed voice assistants' inability to recognize a Black grandmother's accent, they did not perceive this as an instance of unfairness.

In our study, we observed a similar pattern: while most participants acknowledged that gameplay increased their awareness of biases, including previously unknown categories and debiasing methods, the broader issue remains. The fact that ML bias and its impact on underrepresented groups are not adequately recognized, even by those most affected, is concerning. We propose that transformational or serious games can effectively raise awareness of ML biases. These games can serve as a powerful tool to deliberately expose people to ML biases and elevate their consciousness regarding these issues. By providing concrete examples and a vocabulary for discussion — such as the bias categories introduced in our game — these games can become a catalyst for deeper understanding and communication about experiences with ML bias and broadly ML harms [82] and social impact [83].

6.1.2 Collaboration. Previous research has demonstrated that collaboration and social interaction significantly boost student engagement and motivation in learning environments, promoting mutual learning and support [21]. We have integrated this principle into our game mechanism design extensively. For instance, in the 'Discover Bias' phase, we incorporated elements of peer teaching, known to deepen understanding of learning subjects [101]. Participants were encouraged to explain machine learning concepts to one another. Moreover, we noted that dyadic pairs with less prior experience in identifying ML bias engaged in discussions about debiasing strategies as effectively as those with more extensive ML experience. This suggests that dyadic pairing can enable learners to collaboratively develop solutions and make strategic choices, compensating for individual knowledge gaps. This observation merits further investigation. It also raises the question of whether learners with prior ML education may have insufficient exposure to critical issues like bias in ML systems, an area previously noted in the literature [38, 78].

Future research should examine the compatibility of student pairs in the context of ML bias literacy. While prior studies have explored the impact of factors such as skill level, gender, and personality on pair compatibility for optimal learning experiences [15, 49, 50], we advocate extending this exploration to include sensitive identity traits. This is due to the unique nature of ML bias literacy, closely related to vulnerable identities, and the compounded impacts when multiple vulnerable identities intersect [60].

6.1.3 Critical Reflection. Although participants generally rated *Equality Engine* favorably, it is important to acknowledge that certain game mechanics may induce feelings of frustration, discomfort, or other negative experiences due to the nature of our game's content and mechanisms. We contend that these experiences are instrumental in facilitating critical reflection among participants, leading to transformative learning.

Notably, during the game, particularly in the initial 'Discover Bias' phase, participants often express negative feelings. This can be partly attributed to the intimidating complexity of Machine Learning, e.g., participants struggled with identifying bias categories, which has been identified by instructors as a general barrier to teaching non-major students ML [90, 91]. Another potential source of discomfort specifically in our context might stem from the unease of uncovering biases deeply ingrained in their social and cultural contexts, though not directly labeled as negative by participants. This presents a longstanding educational dilemma: introducing vulnerability-related content (e.g., gender, race, class) in learning environments can elicit a range of negative emotional responses from guilt and shame to anger and despair. However, such content also holds the potential to foster constructive dialogue, spur action and allyship, and enhance self-esteem through heightened awareness and mindfulness [96]. The HCI field has similarly explored the concept of 'uncomfortable interaction', intentionally engineering discomfort to create intense, memorable interactions and engage with challenging themes [3]. Halbert and Nathan [41] further argues that design elements provoking negative feelings can challenge users' assumptions and worldviews, fostering critical engagement with these feelings and laying the groundwork for perspective transformation.

In this study, we propose that the initial adverse experiences encountered in the 'Discover Bias' stage of the gameplay a crucial role in engaging participants in profound critical reflection in subsequent phases [80]. As participants advance, particularly during the 'Debias' phase, they are presented with opportunities and scaffolded information, aiding them in processing and contextualizing their initial discomfort. Within this secure gaming environment, participants engage in dialogues, allowing them to articulate, scrutinize, and reshape their perceptions and emotions. They are also encouraged to envision solutions to counter the biases they encounter, potentially mitigating their negative experiences. This process aligns with Mezirow's theory of transformative learning, which asserts that critical reflection on personal assumptions and beliefs is vital for transformative learning [62].

In summary, we recommend that future research incorporate game mechanisms that scaffold critical thinking. While these may inherently be challenging and provoke seemingly counterproductive negative emotions, they are crucial for fostering deeper engagement, reflection, and ultimately, personal and societal transformation.

6.2 Implications for Future Games Aimed at AI/ML Bias Literacy

6.2.1 (Cautious) Adaptability to Wider Audience. Prior research has deeply investigated different modalities to expose diverse stakeholders to various aspects of AI/ML ethics and fairness literature, spanning from toolkit for algorithm developers [83], course materials in computational education classroom [22, 25, 46], to AR-based contextualized dilemma for K12-students [57]. In contrast, our work explores a new and easily-accessible modality, a card-based board game, utilizing bias and debias cards particularly to educate ML bias-debias concepts among post-secondary students (e.g., in the freshman or final year), which, for our targeted group, could serve as complementary means of exposure, outside formal education context.

This being said, our game, 'Equality Engine,' could be adapted for a broader audience. Although it has not been designed or tested for use in a formal classroom setting, it could serve as a fun and engaging supplement to traditional education methods, complementing complex subjects typically covered through lectures (e.g. assignments, and demonstrations on real-world datasets [46]). Furthermore, the game could benefit the general public, beyond our initial target audience, responding to the increasing call to integrate AI/ML fundamentals and ethics into general education [84]. The game mechanics are intentionally simplistic, making them easy to learn without any prior digital or physical gameplay experience. The card contents are free of ML jargon, addressing a gap where many AI/ML ethical educational tools generally require a baseline understanding of technical ML concepts, such as performance metrics and specific algorithms [83]. Given that its structure mirrors general ML pipelines, we believe the game could be further customized for specific audiences to enhance engagement, such as adapting it to contexts like credit scoring for young adults or housing allocation for the homeless.

While this game offers considerable adaptability to a wider audience, it is important to proceed with caution when exposing players to AI/ML bias-related concepts. This is especially critical for future research aimed at expanding the audience to vulnerable groups, such as participants under 18 years of age. This caution underlies the social and cultural sensitivity surrounding job hiring biases, which could evoke a range of reactions, fear, or even anxiety among the audience. To counteract this, we prepared a post-playtest care package, including more information sources on how people combat ML bias, to be shared among the participants to prevent negative feelings, however, due to timing constraints we were unable to test their impact. Future researchers should also apply appropriate aftercare tools to specific audiences. Our post-playtest aftercare intervention cards package can be found in the supplementary material GitHub repository.

Nevertheless, some of our design elements might play a positive role in minimizing the potential negative effect for vulnerable groups. For example, we maintained a collaborative environment, avoiding direct competition and fostering supportive interactions, specifically with two strategies. First, we integrated challenges and a scoring mechanism that encouraged collaboration to optimize scores. For instance, a player might give up immediate points by contributing a ‘debias card’ to a teammate for potential future gains, a tactic frequently discussed during gameplay. Second, we shifted the competitive element outside of playtime and space by comparing scores with previous anonymous players through a leaderboard that displayed the top three scores. This approach provided a challenging goal while minimizing the social anxiety associated with direct confrontation. We observed that players recognized and responded positively to these incentive structures. Peer collaboration in computer science education has enhanced competence, confidence, and persistence, especially among underrepresented groups [42, 58]. Future research could consider these two strategies and explore more when designing collaborative games for vulnerable groups.

6.2.2 Challenges in Outcome Measurement. A specific challenge within ML ethics education is outcome measurement. Most measurements of study outcomes rely on user-generated content and self-reported surveys, which are predisposed to social desirability bias (SDB), and often lack reported evidence of reliability and validity, suggesting potential measurement errors [12, 69]. In our game setting, we encounter similar issues, especially the SDB, which occurs when individuals respond in ways that align with social norms rather than their true feelings, aiming to project a favorable image or avoid negative reactions [95]. The feedback from both surveys and interviews tends to be overly positive, particularly when considering the statistical significance. However, as discussed in our qualitative findings, we observed expressions of confusion and frustration during gameplay. Such data, derived from the learning process rather than the outcome, helped us counteract the bias inherent in outcome measurement challenges. This approach aligns with Brown et al. [12]’s suggestion to include data reflecting the learning process in ethical education settings. Consequently, we recommend that future research explore a process-oriented measurement approach to mitigate SDB, which can persist in outcome-oriented measurements. Additionally, future studies should consider using a control variable scale, employing randomized responses, and utilizing indirect questioning to counteract SDB [95].

6.2.3 Balancing Goals. To effectively engage a targeted audience with a game medium while educating them to increase ML bias literacy, several aspects must be balanced. First, incorporating educational science insights into game design is crucial. As exemplified in our game design, we engaged our audience with real-life scenarios and promoted social interaction and collaboration, reflecting the principles of situated learning [84] and collaborative learning [86] at a meta-level. Future research should consider integrating pedagogical frameworks more deeply in this context, such as employing the Structure of Observed Learning Outcomes (SOLO) taxonomy [90, 91]. Second, we simplified the technological underpinnings of ML, minimizing math and programming content to engage the audience more effectively. While this approach is effective for engagement, it's important to note that more direct methods might better help players grasp ML complexities, such as emphasizing their agency and ability to understand how ML functions [90]. Third, we quantified and oversimplified the effects of various actions into scores to enhance entertainment, rather than accurately representing the impacts of negative biases. Quantitatively comparing the impacts of different biases is inherently challenging. Although no discomfort or negative reactions to this proxy approximation were observed among participants, we advise future research to be aware of potential side effects. Future studies should also consider developing improved methods to depict the effects of various bias countermeasures and promote critical reflection among students.

6.3 Threats to Validity and Limitations

Our study is not perfect. We will discuss threats to our study and the validity of our approach and findings [5, 45, 102]. First, the findings of our study come from a limited and specific sample of post-secondary students. Despite a small sample, the detailed qualitative responses and carefully selected statistical methods — based on the sample distribution and study objectives — demonstrate the rigor of our results. At the same time, as this study is a formative playtest, it is appropriate that we first receive this feedback regarding the playability and legibility of our game with an accessible audience before more broadly recruiting. Furthermore, mixed method moderated study design and data collection facilitated triangulation of results using qualitative and quantitative evidence. Again, our goal was not to share generalizable results, but rather share preliminary findings as a case study of using a transformational game to teach post-secondary students about ML biases. Second, the threat to internal validity comes from unmeasured control variables that could influence the dependent variables. To counteract this threat, we included participants' background information in all regression analyses; however, no significant causality was observed. Third, upon reflection of this work, as it will become publicly available to the general public, we acknowledge that the original design goal of the game might shift meaning, and raise concerns when used outside of the game's original context. Fourth, while our target audience is post-secondary students in general, we are aware of the potential different experiences players may have based on their intersectional identities. For example, our team is embedded in a racially diverse context and is aware that some game elements/wordings such as "helping a student" unintentionally convey an "Ivory tower" or "white savior" message. Our sample was limited in terms of racial and geographic diversity. However, as our game discusses many examples of bias from race to gender to age, our goal was not to have a "perfectly representative sample" but instead to playtest the game. Based on identity experience, we also acknowledge that the game may bring up traumatic memories for participants. To alleviate this unintentional trauma, we incorporated post-playtest aftercare interventional cards for self-care, which consist of strategies one can apply to self-help or seek support from the community and advocacy groups and network with them. In future game design iterations, we hope to expand and test this game with a wider audience, after revising or removing those unintentional messages. Lastly, one final limitation comes from participants' self-reported increase in ML bias knowledge, as they were not tested on ML bias concepts and correctness per se. Self-report findings can only represent participants' perception of knowledge increase but not actual knowledge. Moreover, all self-report pre-post test survey scales were developed by the research team by taking inspiration from prior work, containing a small set of items, resulting in less reliable responses for some measures. In the future, we recommend researchers use validated scales for pre-post tests.

7 Conclusions

This paper discusses the design, development, and mixed-method assessment of a novel online board game *Equality Engine* to teach players specifically about ML biases and debiasing techniques. As a case study, this game was playtested with post-secondary students. A focus group formative playtest with six pairs (N = 12) was conducted. Participants' self-reported data indicate that after just three rounds of gameplay, the game has significantly increased their perceived knowledge about various ML bias concepts regardless of ML background experience. The game provided a deep exposure to new ML concepts, and participants collaboratively learned through engaging in meaningful conversations and critical reflections. The game was perceived as useful, easy to use, and enjoyable. Participants found the game easy to use because of the social aspect and immersive ability, and they would use the game in the future because of the knowledge they gained from playing it. We hope that this game will empower students and other professionals, augment their ML bias/debias knowledge, help them learn about ML bias detection and prevention techniques, and engage them in important critical discourse about emerging ML technologies and the ethical dilemmas they pose.

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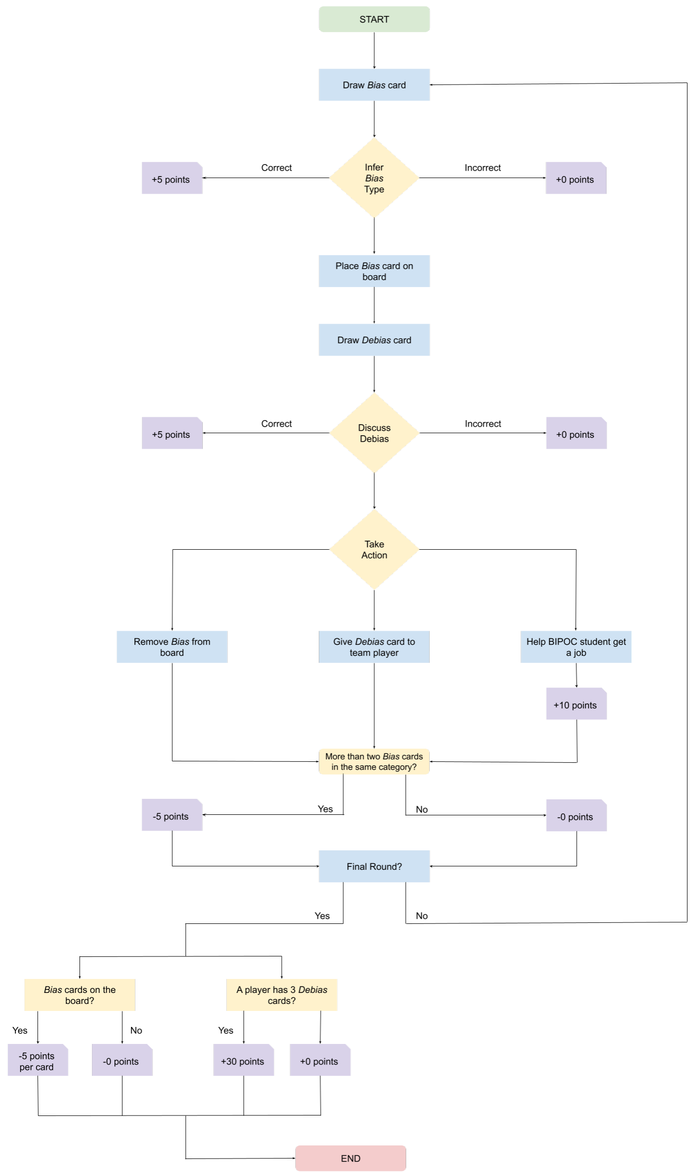

Fig. 6. High-level overview of the *Equality Engine* game workflow.

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A Appendix: Game flow chart, ML bias and debias examples

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Table 2. Summary table showing five types of ML biases, each with its definition, and one example card along with its source. The context of most bias i.e., subtitle and content, represents hiring context except for facial recognition bias example, which can be generic as well.

Bias Name	**Definition**	**Card Subtitle**	**Card Content**	**Reference**
Representation Bias	*"Representation bias occurs when the development sample underrepresents some part of the population, and subsequently fails to generalize well for a subset of the use population."* [94]	Facial Recognition Bias	If using mostly photos of white men to train the machine to learn human facial recognition, the machine will perform poorly on recognizing other groups, such as people of color and women.	[73]
Modeling Bias	*"Modeling bias refers to biases that emerge during the training of a model, often resulting from factors like the evaluation methods used or the choice of metrics"* [94]	One-size-fit-all	In reality, one single model might not work for people with different backgrounds. For example, a given variable, such as GPA, can mean something quite different across groups.	[73]
Historical Bias	*"Historical bias arises even if data is perfectly measured and sampled, if the world as it is or was leads to a model that produces harmful outcomes"* [94]	Inherited Proportion	Amazon used an algorithm to aid in their hiring process which resulted in disproportionately selecting men over women due to the fact that they had hired more men in the past.	[51]
Measurement Bias	*"Measurement bias occurs when choosing, collecting, or computing features and labels to use in a prediction problem."* [94]	Persistency	Employers tend to discriminate against women and ethnic minorities, and a recent analysis suggests that little has been improved over the past 25 years.	[104]
Deploy Bias	*"Deployment bias arises when there is a mismatch between the problem a model is intended to solve and the way in which it is actually used."* [94]	Problematic Generalization	Some algorithms were pre-built with general assessments from general dataset, which may not adapt well to a specific job in a specific company.	[73]

Table 3. Summary table showing five ML debiasing techniques with content and sources.

Debias Title	**Content**	**Questions in the card**	**Reference**
Consult Experts	Seeking domain experts' advice would be helpful!	What experts do you think you can turn to for help in this situation?	[73]
Audit the algorithms	The city of New York assembled Automated Decision Systems Task Force to recommend a process for reviewing the city's use of algorithms.	What else do you think government could do to promote the fairness in ML hiring context.	[30]
Checking team's own biases	The development team could check their own biases, which might creep into the algorithms.	How can teammates help each other find their bias?	[54]
Protected Attributes	Remove protected attributes (or demographic characteristics) from the database might be helpful.	What attributes may be affected by bias and could be removed?	[24]
Penalize Bias	Usually, under-represented samples in the data are neglected since they do not influence the overall accuracy as much. Model developers could add some terms to penalize such biased predictions.	In job application scenarios, what kind of bias for BIPOC students might occur and developers should pay attention to?	[24]
Swapping	To remove gender bias in the dataset, male entities could be swapped for female entities and vice-versa.	How can this method be applied to address other forms of bias?	[24]